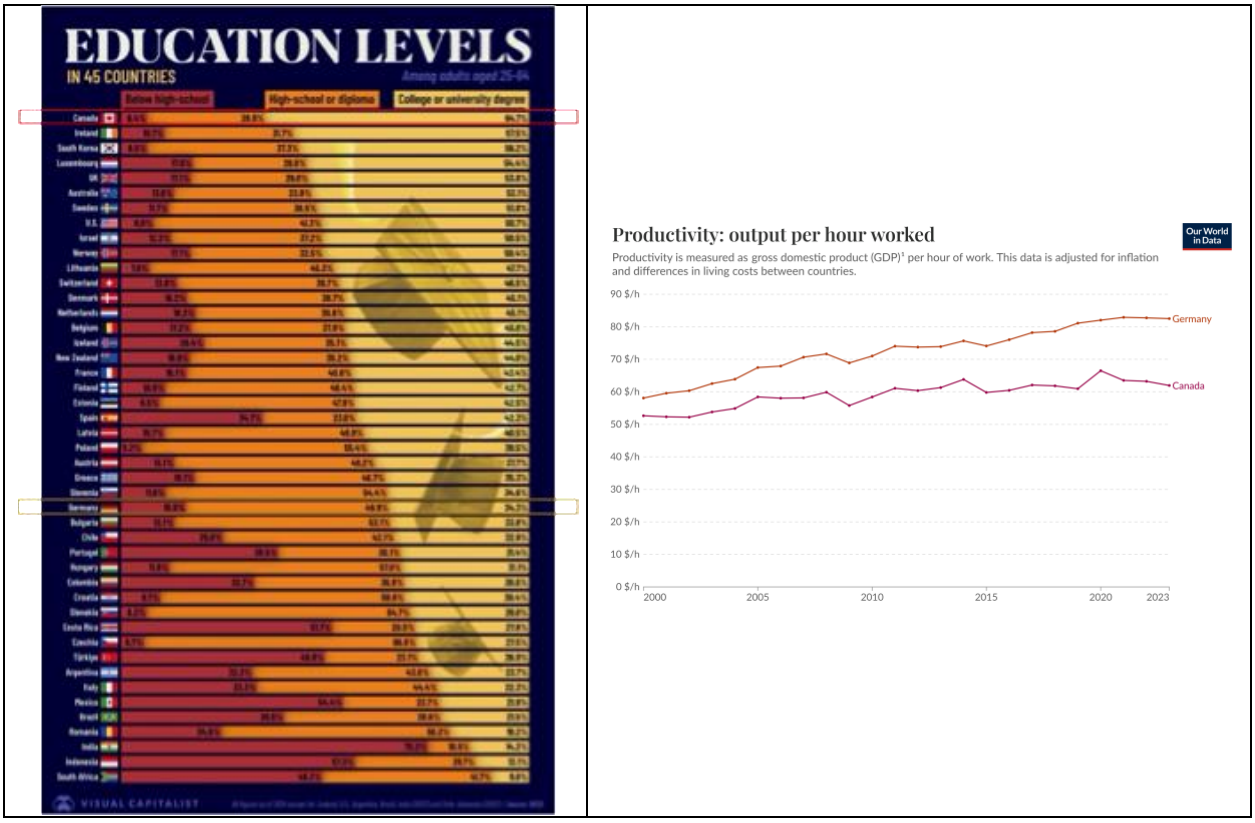
Education and productivity

Consider the two charts provided below. The chart on the left shows education levels across the 45 OECD countries with boxes highlight two countries, Canada and Germany. Canada tops the charts for educational attainment of its citizens, with nearly 2/3 of working age adults holding a university degree. Germany has barely half that number, at 34.3%. The chart on the right shows productivity levels for the same two countries over a nearly 25 year period from 2000 to 2023. In 2023, Germany enjoyed productivity levels that are over 30% higher than Canada's, a gap that has grown substantially over the period highlighted.



1. Juxtaposed, the two charts present a puzzle relative to the way we have discussed productivity in class. Define the puzzle as clearly as you can using concepts from class. For this question, you don't have to explain why the puzzle exists, i.e., what is causing it -- just explain what the puzzle is. Make sure to refer to the data provided and reference both highlighted countries. (Word Count Guidance: 100). 10 marks.

2. Now, focus on Canada. Develop three plausible hypotheses to explain why Canada's strong educational attainment has not translated into faster productivity growth. In your answer, refer explicitly to both charts—education and productivity—and connect your reasoning to relevant growth and productivity concepts from the GMM framework. You should draw on your knowledge of Canada gained from our class discussions. Your answer will be judged on the plausibility, importance and clarity of your hypotheses, as well as the connections you make to the data provided in the two charts. Make sure your three hypotheses are distinct from each other – don't give minor variations of the same hypothesis (Word Count Guidance: ~150 words; 3 x 50). 15 marks = 3 x 5 for each hypothesis.

3. Now, focus on Germany. It is often said—humorously but accurately—that Germany's economy is built around firms that "make the thing that makes the thing that goes in the thing that is the thing." (Think: companies that produce machines that make components—like bearings or sensors—that go into finished products such as vehicles or industrial equipment.) Drawing on this insight, explain how Germany's economic structure might help account for the pattern shown in the two charts. Again, be sure to link the two pieces of data provided in the charts and focus your answer completely on Germany (Word Count Guidance: ~100 words). 10 marks

4. The Government of Canada is preparing its Fall Budget, which—for the first time—will clearly distinguish day-to-day operating spending from long-term capital investment. The new framework aims to prioritize projects that generate lasting economic benefits for Canadians—major resource developments (mines, pipelines), housing, clean-energy infrastructure, transportation networks (railways, ports), and other "generational investments" designed to boost growth and attract domestic and foreign investment.

   Now imagine that Prime Minister Mark Carney visits Ivey the week before the budget is released, and you are granted a five-minute one-on-one meeting with him. He wants your candid view on whether Canada's education system is well aligned to support this new wave of transformative investment—and your ideas for improving it. Carney seeks your opinion on a specific budget proposal he is considering: to cut funding for university research. He tells you:

   > "I'm concerned that our universities aren't delivering clear productivity benefits, so I hesitate to keep pouring money into something that may not be working. Our fiscal outlook is increasingly difficult—deficits are rising, and so is our debt-to-GDP ratio. The proposal on the table is to reduce spending on basic scientific research. My view is this: let other countries do the discovery work—scientific knowledge spreads quickly across borders anyway. Why spend scarce resources on efforts that benefit others as much as us? Given our fiscal constraints and strategic challenges, we should focus on applying new ideas in our industries, not discovering them—be the best at adopting innovations developed elsewhere."

As the starting point for your answer, please state either **"Yes, good idea"** or **"No, bad idea,"** then explain your reasoning. (Word Count Guidance: ~150 words). 15 marks

Model Answer — Question 1

The charts reveal a clear puzzle: Canada leads the OECD in educational attainment, with nearly two-thirds of adults holding a tertiary degree, yet it lags Germany in productivity per hour worked. Germany ranks much lower—only about one-third of its population has an advanced degree—but achieves both higher productivity levels and faster productivity growth.

Within the GMM growth framework, education (human capital) is a key driver of productivity. Yet Canada's experience suggests that high educational attainment alone does not automatically translate into higher productivity. Conversely, Germany's strong performance implies that factors other than formal education levels may be more important in yielding superior productivity outcomes.

Model Answer — Question 2

Weak capital deepening and investment incentives

Canadian firms invest less in machinery, equipment, and digital infrastructure than international peers, leaving educated workers without the tools to boost productivity. Human capital alone cannot raise output without complementary capital and innovation. Weak incentives for business investment—high costs, uncertainty, and limited competition—have slowed capital deepening and productivity growth.

Industrial structure and resource dependence

Canada's economy is heavily concentrated in resource extraction and low-productivity services, where extra education yields limited returns. Output depends more on natural endowments than workforce skills, so rising educational attainment has little effect on overall efficiency. With a small base of knowledge-intensive industries, Canada's educated labour force remains underutilized and productivity growth subdued.

Weak diffusion and adoption of innovation

Canadian firms lag in adopting frontier technologies and management practices, creating weak productivity growth. High educational attainment generates potential human capital, but limited innovation ecosystems and slow diffusion of best practices prevent it from translating into aggregate productivity improvements.

Weak competition and limited scale economies

Many Canadian firms operate in small domestic markets protected by limited competition. Even with a well-educated workforce, firms may face little pressure to innovate, adopt new technologies, or expand internationally. In the GMM framework, weak competitive dynamics reduce productivity growth by slowing creative destruction and diffusion of best practices.

Skill mismatch and overqualification

Canada's education system produces many graduates whose skills don't align with the jobs available in the economy. High formal attainment may hide underemployment or weak absorption of

advanced skills. When highly educated workers perform routine or service-sector tasks, average productivity gains remain limited.

Fragmented coordination between education and industry

Educational institutions and employers may be poorly aligned in Canada. Curricula are often developed independently of industrial needs, limiting the translation of learning into productive capability. In contrast, countries with stronger institutional linkages between training systems and firms can convert education more directly into output efficiency.

Model Answer — Question 3

Germany's economy is described as one that "makes the thing that makes the thing," suggesting a deep network of firms producing the machines, tools, and components used by other producers. The educational chart shows that, while relatively few Germans hold university degrees, a large share have completed high-school or diploma-level education. This pattern implies a workforce with strong technical and applied skills—well matched to a manufacturing- and engineering-based economy.

The productivity chart shows that this structure yields high output per hour worked: education in Germany appears tightly connected to practical production needs, allowing technical training to translate directly into efficiency and innovation on the shop floor. Together, these features help explain Germany's strong productivity despite lower formal tertiary attainment.

Model Answer — Question 4

Yes, good idea.

Given mounting deficits and a rising debt-to-GDP ratio, the government must focus scarce resources where they deliver the highest short-term productivity payoff. Basic research is vital in principle, but in practice Canada's challenge lies less in discovery than in diffusion and execution—turning known ideas into output. Redirecting funds from long-horizon science toward applied partnerships, commercialization support, and workforce skills would yield faster returns and strengthen fiscal credibility. Since scientific knowledge spills across borders, Canada can benefit from discoveries made elsewhere without bearing the full cost. A pause or rebalancing of basic-research funding could free resources for infrastructure, clean energy, and industrial modernization—the "generational investments" most likely to raise growth and tax revenues. To avoid long-term damage, however, such cuts should be clearly temporary and paired with metrics to ensure that future surpluses can rebuild the research base once fiscal conditions improve.

No, bad idea.

Fiscal pressures are real, but cutting university research would weaken Canada's long-run growth capacity—the very thing needed to stabilize debt ratios over time. Basic science is the seed corn of innovation: it builds the absorptive capacity that allows firms to adapt, refine, and commercialize

foreign discoveries. Without it, Canada risks dependence on imported technologies, brain drain, and a shrinking ecosystem for advanced industries. Moreover, research spending is a capital investment, not just a cost—it raises future productivity and revenues, improving fiscal sustainability. The issue isn't that research "isn't working," but that Canada underinvests in connecting labs to markets. A smarter path is to maintain core funding for discovery while tightening incentives for collaboration, proof-of-concept work, and applied diffusion. Fiscal consolidation should come from routine operating expenditures, not from cutting the knowledge base that underpins long-term competitiveness and the very capacity to service future debt.